mitrafield

Ask a question.
Get a narrated explainer video.

MitraField is an AI studio that turns a question, a lesson plan or a PDF into a finished explainer video: a script, the right picture for every scene, narration in your own cloned voice, music and subtitles. Numbers, maps and math are drawn from real data, so the video is correct, not just pretty.

Google Cloud AI Builder Cup 2026
Theme: **Media, Content & Digital Experiences**

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Live: **mitrafield-aibc26.web.app** · github.com/ViraSawad1021/MitraField

What it is, and how it meets each judging criterion.

TECHNICAL MERIT & GEN AI · 40%

Google AI models working as a team

Gemini writes, plans, codes, reviews and fact-checks; Nano Banana, Veo, Gemini TTS and Lyria make the media. Every step checks the one before it. Deployed on Cloud Run with a durable job queue, a GPU voice service and automatic, gated releases.

PROBLEM ALIGNMENT & IMPACT · 25%

Correct explainers in minutes, in many languages

A good explainer takes a studio days. AI video tools are fast but get facts wrong. MitraField makes a correct, narrated video in minutes for about one to three dollars, and translates it in one click, for teachers, NGOs, newsrooms and creators.

INNOVATION & CREATIVITY · 25%

The right kind of picture for every scene

Instead of one video model for everything, an AI director picks one of eleven kinds of visual per scene: real math animation, real maps, charts, timelines, or AI imagery only where it helps. Plus your own cloned voice and an invisible watermark anyone can verify.

USER EXPERIENCE & DESIGN · 10%

You approve the plan; change any scene in plain words

Nothing is spent until you approve the storyboard. Every screen has one clear next step, any scene can be changed by asking ("make it simpler"), AI pictures are labelled, cost is shown per video, and it works on a phone.

THE PROBLEM Problem alignment

Video is how people learn now.
Making a correct explainer is still slow, costly and English-only.

days

of work per minute of video

Script, research, animation, maps, charts, narration and editing. Teachers, NGOs and small creators cannot afford a studio or an editor.

wrong

facts from AI video tools

Text-to-video models make beautiful clips but invent coastlines, garble numbers and draw equations as squiggles. In a lesson, a wrong number is worse than no video.

one

language, one format

Most explainers exist only in English and only long-form. A Hindi, Nepali or Bahasa version, or a vertical Short, means starting again.

WHO IT IS FOR Potential impact

Correct explainers for anyone who teaches.
In their language, in their format.

Teachers and schools

Paste a lesson plan, get a narrated video with accurate diagrams, then a vertical Short for revision. One teacher, minutes, a few dollars.

NGOs and newsrooms

Explain a policy, a disease or a crisis with real maps and checked numbers, and publish it in several JAPAC languages the same day.

Creators and educators online

Run an explainer channel in your own voice, with the same presenter and style in every episode, long-form and Shorts from one place.

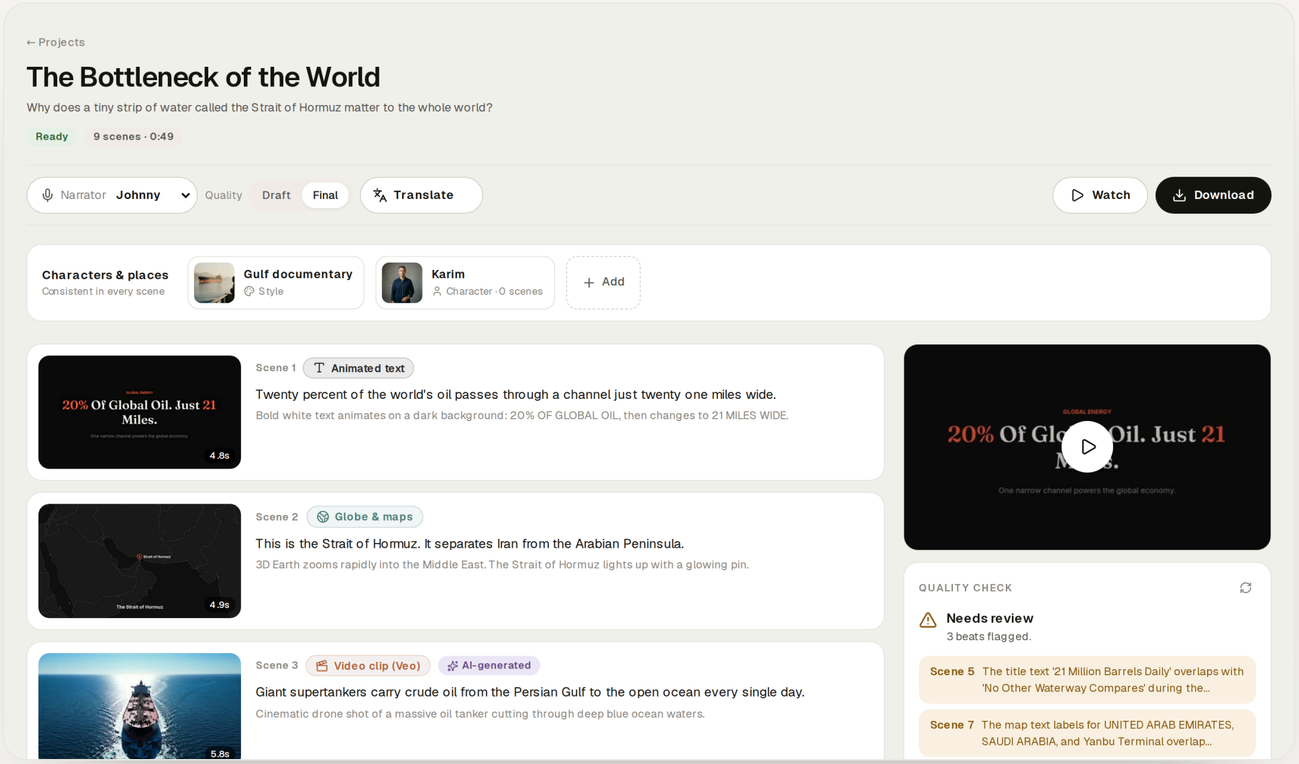
Against misinformation: every number, map and equation is computed from data the user can see, then fact-checked against their sources and Google Search.

Nine narration languages: English, Hindi, Nepali, Japanese, Indonesian, Korean, Spanish, Tamil and Vietnamese.

THE SOLUTION

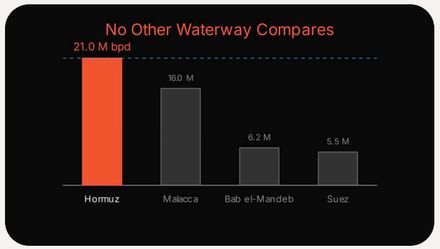
One workspace from a question to a finished video.

- ✓ **Ask** a question, or paste notes, a lesson plan or a PDF.
- ✓ **Review the storyboard** the AI director writes: the script, scene by scene, and the picture chosen for each one. Nothing is spent before you approve.
- ✓ **Render**: every scene is narrated and drawn to the exact length of its sentence, then checked for quality and facts.
- ✓ **Change any scene** by editing it or asking: "simpler", "show it on a map". Only that scene is redone.
- ✓ **Download** an MP4 with music and subtitles, or make a Hindi or Nepali version in one click.



THE CORE IDEA Innovation

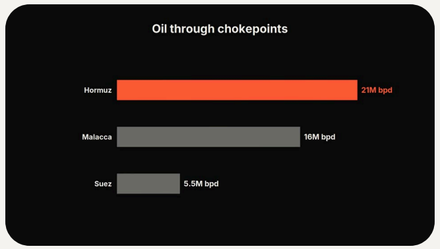
The right kind of picture for every scene. Real data where facts matter.



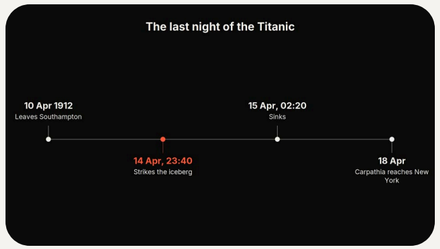
Math animation
Real equations and graphs, drawn by code



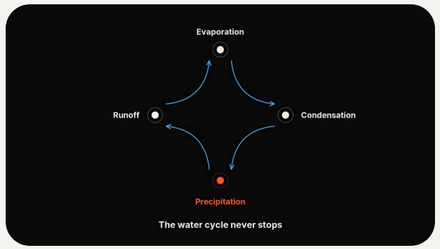
Globe & maps
Real country borders, routes and pins



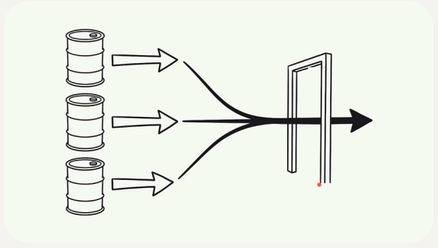
Data chart
Bars and trends from your numbers



Timeline
Events placed at their real dates



Flow diagram
Steps, cycles and cause and effect



Whiteboard
A sketch drawn stroke by stroke



Animated text
Key numbers and definitions



Cinematic shot AI
Places, history, metaphor



Presenter AI
One person, same face every time



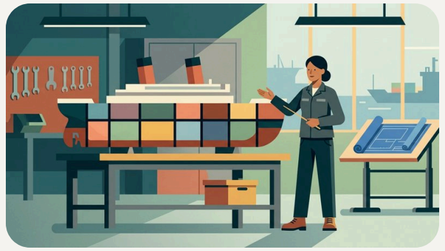
Video clip (Veo) AI
Real motion, at most two per video

Eleven kinds of visual (an animated illustration is the eleventh). The director picks one per scene. Only the three marked AI are generated pictures, and the app labels them "AI-generated" on every scene. In our 12 finished videos, 62% of scenes were drawn from data and code.

STYLE AND CONSISTENCY Creativity

Ten looks. The same ship, and the same person, in every shot.


Cinematic


Editorial flat


Premium 3D


Ink & wash


Woodblock, and 5 more

With the storyboard, the director defines the recurring characters, objects and places. Each gets one reference image, and every AI picture and Veo clip is drawn from it.



RMS Titanic
object, defined once



Thomas, the presenter
a 1912 shipbuilder


Scene 1 · still

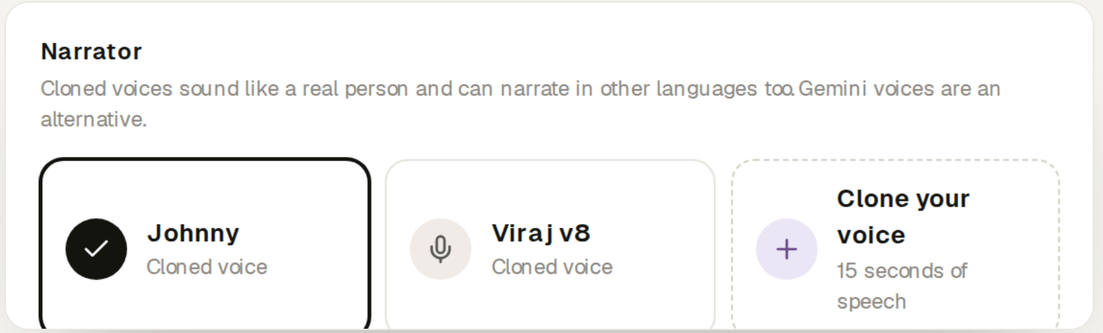

Scene 3 · Veo clip


Scene 7 · Veo clip

VOICE CLONING Innovation

Narrated in your own voice.
Recorded once, in about 15 seconds.

- ✓ Record or upload 3 to 20 seconds of clear speech on the Voices page. Gemini transcribes it if you do not type what you said.
- ✓ The voice is cloned once with OmniVoice, an open model we run on a Cloud Run GPU (NVIDIA L4) that switches off when nobody is using it.
- ✓ The same voice narrates every video, including translations: our Hindi demo is narrated by the same cloned voice as the English one.
- ✓ Consent is required: the person confirms the voice is theirs or used with their permission, and samples are never public.
- ✓ If the GPU is unavailable, narration falls back to a Gemini voice, so a video never fails because of it.



Choosing a narrator: the cloned voices come first, with "Clone your voice" one click away. Gemini voices remain an option for more languages.

HOW GOOGLE AI IS USED Technical merit · meaningful use of Gen AI

Each model does the job it is best at.

JOB	GOOGLE MODEL	WHAT IT DOES IN MITRAFIELD
Director	Gemini 3.1 Pro	Reads the question and sources, writes the script scene by scene, picks the visual for each scene, defines the characters and places.
Code and planning	Gemini 3.5 Flash	Writes the math-animation code, fills in maps, charts and timelines from the sources, rewrites one scene on request.
Pictures	Nano Banana (Gemini 3.1 Flash Image)	Character and place references, cinematic stills, whiteboard sketches, all drawn in the chosen look.
Video	Veo 3.1 Fast (Veo 3.1 Lite for drafts)	At most two hero clips per video, starting from an on-model first frame.
Narration and music	Gemini TTS · Lyria 3	Gemini voices in nine languages (cloned voices run on our GPU service); a music bed made for the topic, kept low under the voice.
Checking	Gemini with Google Search grounding	Reviews the frames of every finished scene and checks every claim against the sources and Google Search, with links.

NOT ONE PROMPT Technical merit · implementation

Seven steps that check each other.

01

Choose the visual

The director picks per scene under rules: facts go to data and code, Veo at most twice, your instructions come first.

02

Write and repair code

Gemini writes the animation code; if it fails, the error goes back to Gemini until it runs.

03

Look and fix

Gemini looks at the rendered frames for overlapping or cut-off text and asks for a corrected scene.

04

Listen and retake

Every narration line is transcribed and recorded again if words were dropped, added or garbled.

05

Stay consistent

Reference images feed every AI picture and Veo clip, so people and objects look the same throughout.

06

Review the finished video

Gemini reviews every scene, and length, black or frozen frames, silence and loudness are measured. One click fixes a flagged scene.

07

Check the facts

Claims are checked against the sources and Google Search; wrong ones go back to the director with the evidence.

Never a broken video

A refused Veo clip becomes a still, a failed scene becomes animated text, and a missing GPU voice becomes a Gemini voice.

TRUST AND SAFETY Responsible use of Gen AI

Checked before you see it. Provable after you share it.

Fact check

Every number and claim, spoken or on screen, checked against your sources, then Google Search, with links to the evidence.

Invisible watermark

Every export carries a mark you cannot see. Anyone can upload a copy on the Verify page; it survives re-uploading, compression, resizing and trimming.

AI labels

Scenes with AI-generated pictures are labelled in the app. Veo and Nano Banana output also carries Google's own SynthID.

No deepfakes

Requests to put words in a real person's mouth are refused. Real people in picture prompts become anonymous stand-ins.

Safe with uploads

Uploaded PDFs and notes are treated as data. A note saying "ignore all previous instructions" was ignored while its real facts were used.

Sandboxed code

Code written by the model runs in a sandbox with no network access.

Spending limits

Every model call is priced and logged. Hard caps of $25 a day and $5 a video, a passcode on anything that spends, and a cheaper draft mode.

Private by default

Uploaded sources and voice samples are never public. Videos sent to Verify are deleted right after they are checked.

BUILT ON GOOGLE CLOUD Technical merit · scalability

Everything runs on Google Cloud, and scales to zero when idle.

Firebase Hosting

Serves the web app worldwide from a CDN and passes API calls to Cloud Run

Cloud Run · API

Public and stateless. Checks the passcode and spending caps, saves projects, queues the work. Never renders.

→

Cloud Tasks

A durable job queue: retries with backoff, rate limits, and jobs survive restarts and new releases

Cloud Run · worker (private)

Writes, narrates and renders one scene per job, up to six at once; assembles, watermarks, runs the quality and fact checks

Firestore · Cloud Storage · Secret Manager

Project state with safe concurrent updates · videos and images · API and watermark keys

→

Gemini API

Gemini, Nano Banana, Veo, TTS, Lyria, Google Search grounding

Cloud Run GPU (NVIDIA L4)

Cloned-voice narration; private, off when idle

Cloud Build · Artifact Registry

Container builds and images

Cloud Billing budget

Alert on top of the in-app caps

Each service has its own minimal-permission account; the public API cannot call models or render, and only the worker can read the Gemini key.

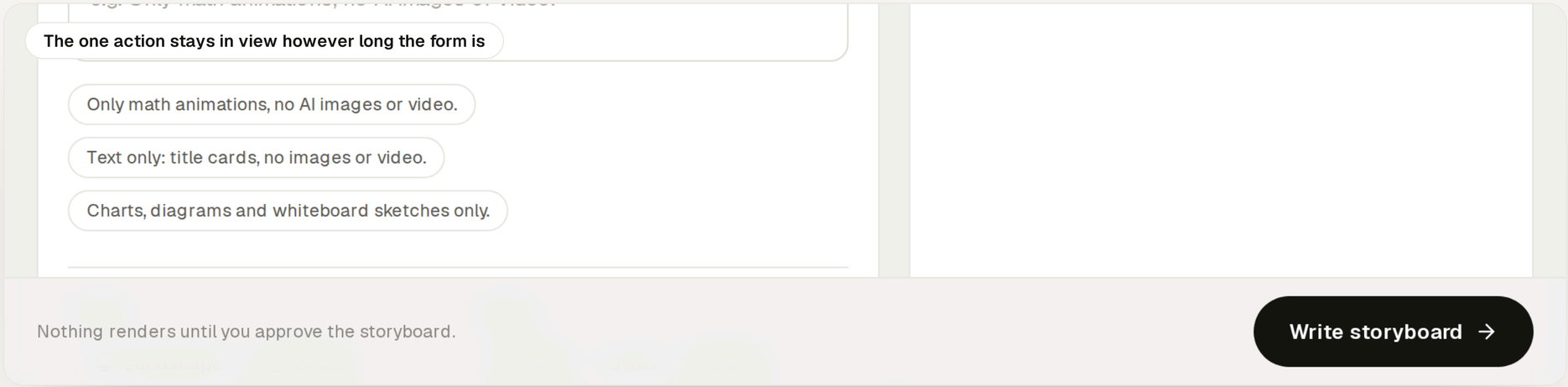
SCALABILITY AND SUSTAINABILITY Technical merit

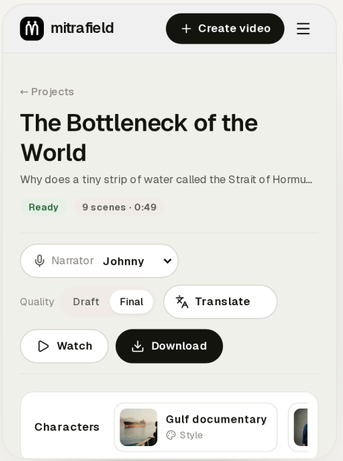
Cheap to run, ready to grow.

AREA	HOW IT WORKS TODAY	NEXT
Scale	The API and the worker scale on Cloud Run from zero to many instances; the queue spreads scenes across workers, six at a time by default	Raise the worker limit and the queue rate; one setting each
Cost when idle	Every service, including the GPU, scales to zero; idle cost is storage only	Keep one warm GPU only during busy hours
Cost per video	About $0.80 to $2.80 in model calls in our runs, shown in the app scene by scene; editing re-renders only the changed scenes	Batch pricing for final renders, per-user budgets
Reliability	126 automated tests on every change; releases go out as a test copy first and only go live after a real check (database, a model call, the watermark and the GPU voice)	Load tests and per-tenant queues
Extending	A new kind of visual is one file; the director discovers it automatically	Quote cards, code walkthroughs, archival footage

Simple for a teacher. Deep control when you want it.

- ✓ **One next step on every screen:** Write storyboard, Render video, Update video, Download.
- ✓ **You approve before anything is spent:** the storyboard comes first, rendering second.
- ✓ **Edit in plain words:** "simpler for a 12-year-old", "show it on a map". Only that scene changes.
- ✓ **Instructions for the whole video:** e.g. "only math animations, no AI images".
- ✓ **Accessible output:** subtitles with every video, loudness set for streaming, nine narration languages.
- ✓ **Landscape or vertical Short** from the same workspace.
- ✓ **Honest by design:** AI pictures labelled, cost shown per video, quality and fact checks visible.
- ✓ **Works on a phone**, and respects reduced-motion settings.





RESULTS FROM THE PROTOTYPE

Finished explainers, each from a single prompt.

12

finished videos: geography, math, biology, history, technology, acoustics, earth science

3

languages so far (English, Hindi, Nepali), 3 looks, landscape and a vertical Short

62%

of 116 scenes drawn from data and code, so their facts were computed, not generated

4–9 min

to render a one-minute video, unattended, with no manual edits



Woodblock look · how a tsunami forms · $1.18



Vertical Short · vaccines, with a recurring presenter



Hindi version · labels redrawn in Devanagari, same cloned voice

WHAT IS DIFFERENT Originality

Beautiful is easy. Correct is the product.

	TEXT-TO-VIDEO TOOLS	MITRAFIELD
Numbers on screen	Drawn by an image model, often wrong	**Computed and typeset by code**
Maps	Approximate coastlines, invented borders	**Real border data, exact places**
Math and graphs	Pretty squiggles	**Real animation of the actual function**
Length	A few seconds per clip	**Full narrated explainers, 1 to 6 minutes**
Changing one moment	Generate the whole clip again	**Edit or redo a single scene**
Narrator	A stock voice, or none	**Your own cloned voice**
Languages	Prompt again and hope	**Same visuals, new narration, one click**
Checking and proof	None	**Fact check, quality check, invisible watermark**

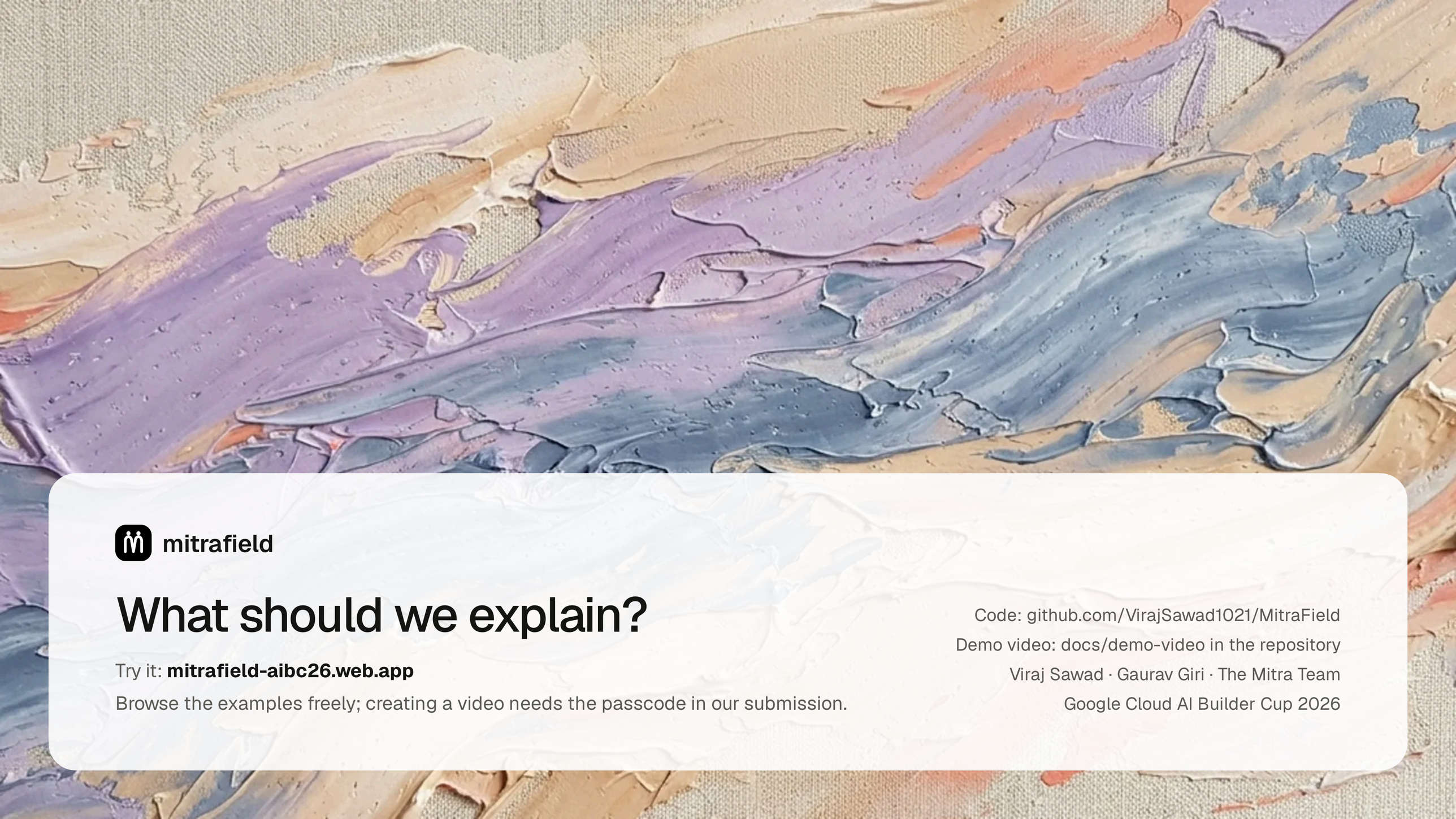
What we would build next.

Limits today

- ✓ No user accounts yet: browsing is public, creating needs a shared passcode.
- ✓ The first cloned-voice line after a quiet period waits 20 to 45 seconds while the GPU starts.
- ✓ Fact checks greatly reduce errors but cannot promise every claim is right; flagged claims are shown, not hidden.
- ✓ Heavy cropping or deliberate editing can remove the watermark.

Next

- ✓ Sign-in with Firebase Authentication and private workspaces for schools and teams.
- ✓ Learn mode: a short quiz after each video, with results in BigQuery to show where learners struggle.
- ✓ More languages and regional voices, with glossaries for subject terms.
- ✓ Publishing straight to YouTube and learning platforms.



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What should we explain?

Try it: **mitrafield-aibc26.web.app**

Browse the examples freely; creating a video needs the passcode in our submission.

Code: github.com/ViraSawad1021/MitraField

Demo video: docs/demo-video in the repository

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